

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) |  |  | <p><b>Indicative content</b></p> <ul style="list-style-type: none"><li>• addition of acid to form soluble salt</li><li>• excess acid to ensure maximum amount of magnesium salt</li><li>• addition of soluble carbonate</li><li>• excess to ensure all magnesium ions precipitated</li><li>• filter and dry precipitate</li><li>• appropriate equations</li></ul> <p><b>5-6 marks</b><br/>Full description of two stages; good attempt at two equations<br/><i>The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i></p> <p><b>3-4 marks</b><br/>Full description of one stage; attempt at one equation<br/><i>The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.</i></p> <p><b>1-2 marks</b><br/>Gives reactants or partial description of one stage<br/><i>The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give an answer worthy of credit.</i></p> |                 | 4   | 2   | 6     |       | 6    |
|          |     |  |  | Question 12 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0               | 8   | 6   | 14    | 3     | 11   |